

# VOLLEY Gen5 | Final-Year Project Report

Computational design, evidence review, and engineering disposition

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# Contents

|                                                  |           |
|--------------------------------------------------|-----------|
| <b>Abstract</b>                                  | <b>1</b>  |
| <b>Document control and evidence convention</b>  | <b>1</b>  |
| <b>Executive summary</b>                         | <b>2</b>  |
| <b>1. Introduction</b>                           | <b>3</b>  |
| <b>2. Literature review</b>                      | <b>4</b>  |
| <b>3. Research gap</b>                           | <b>5</b>  |
| <b>4. Objectives and outcome</b>                 | <b>5</b>  |
| <b>5. Research methodology</b>                   | <b>5</b>  |
| <b>6. Timeline</b>                               | <b>6</b>  |
| <b>7. Work progress</b>                          | <b>7</b>  |
| <b>8. Results</b>                                | <b>7</b>  |
| <b>9. Conclusion</b>                             | <b>10</b> |
| <b>10. Future work</b>                           | <b>12</b> |
| <b>11. References and project records</b>        | <b>12</b> |
| <b>Appendix A. Numerical and CAD audit trail</b> | <b>12</b> |
| <b>Appendix B. Submission and freeze status</b>  | <b>13</b> |

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**Scope of this report.** VOLLEY is a system-level engineering design and computational research project. The Gen5 analysis, comparisons and decision record are complete for the stated models. The results are predictions, with numerical cross-checks where identified. Physical prototype testing, CubeSat qualification and provider accommodation belong to a later development phase. Gen6 is presented separately as future work.

## Abstract

This final-year project report evaluates a magazine-fed electromagnetic CubeSat deployer as a *computational design study*. For a twelve-payload 3U configuration, the coupled model predicts a 16.029 m/s rated release at 10.068 g and 2.782 kJ gross shot draw. An independent Cartesian two-body calculation reproduces a 28.800775 km immediate semi-major-axis rise for the rated 450 km case. A native FreeCAD 1.0 review document and STEP exports expose an 11 mm transverse shortfall in the specified side-fed layout, with 32,915 mm<sup>3</sup> of exact-solid overlap per cassette. The estimated 126.6 kg dry mass gives 10.547 kg of deployer per carried 3U, failing both the 2 kg economic screen and an approximately 6 kg spring-canister parity comparison. Atmospheric lifetime, exact payload interfaces and full mission delivery remain conditional or open. There is no physical prototype or VOLLEY test data. The completed result is a documented model-based decision against this Gen5 3U configuration, with Gen6 scale-up treated as future research.

**Keywords:** CubeSat deployment; electromagnetic launcher; systems engineering; numerical verification; CAD interference; orbital mechanics; model credibility.

## Document control and evidence convention

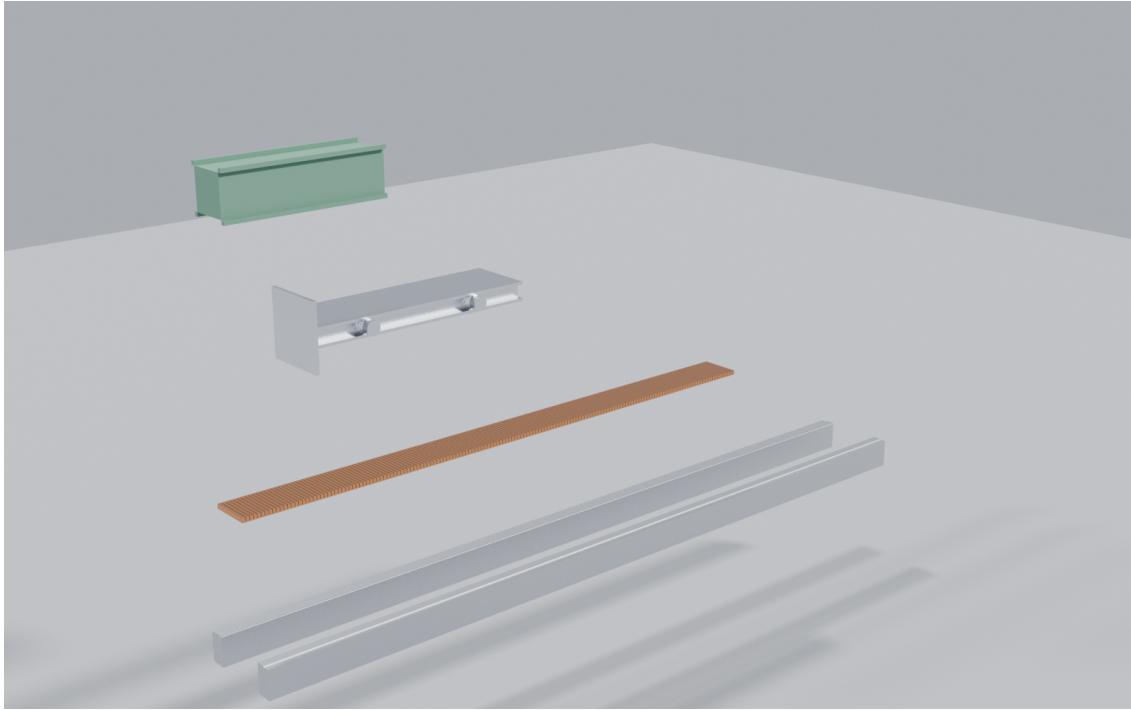


Figure 1: Analysed Gen5 electromagnetic deployer configuration. CAD render; no physical article exists.

| Item          | Review baseline                                                                                                    |
|---------------|--------------------------------------------------------------------------------------------------------------------|
| Configuration | Gen5 twelve-3U side-fed reference; 1.5 m track, 1.3 m powered stroke                                               |
| Report status | Academic review draft; negative criteria retained                                                                  |
| Evidence      | Model outputs, selected independent numerical checks, native CAD import/export and prior literature                |
| Exclusions    | Hardware measurements, payload qualification, provider ICD approval, lifetime cross-check at the exact rated point |
| Build record  | Reproduce PDF from this Markdown with Pandoc and XeLaTeX; inspect local source and run sheets                      |

The supplied college documents define final-review criteria and a slide example. They do not supply a mandatory report template. This report therefore uses a conventional chapter structure; title/signature pages and institution-specific declarations should be added only from the official template. No unperformed test, approval or authorship certification is implied.

## Executive summary

VOLLEY investigates a last-mile problem in rideshare launches: a secondary CubeSat reaches orbit but ordinarily receives the orbit selected for the primary payload. A spring deployer separates it safely, often at roughly 1–2 m/s; that relative impulse, later drag differences, or host maneuvering can change its in-track phase. Deployment time by itself does not create persistent phase separation from an otherwise unchanged co-orbital host. A larger, deliberately commanded release velocity can also change orbital energy. The thesis evaluates whether that added capability justifies a complete deployer system. The mission model splits responsibilities between a maneuvering host, which performs coarse repositioning, and a deployer, which supplies each satellite’s commanded release condition.

This report analyses one Gen5 solution: a magazine-fed, linear synchronous motor driving a reusable magnetic sled. The rated 3U model predicts **16.0 m/s** exit speed at **10.07 g**. Its modeled dry-mass rollup is **126.6 kg**, and the power model predicts **2.782 kJ** gross electrical draw per shot. A closed-loop simulation predicts **0.0274 m/s** ( $3\sigma$ ) exit-speed dispersion at a 15.8 m/s setpoint. In a stated orbital case, a maximum-speed release raises semi-major axis by **28.8 km**, with a **1.60x** modelled lifetime multiplier at mean solar activity. The lifetime result is atmosphere-sensitive and has not been independently rerun at the exact current operating point.

Gen5 also produced a decisive negative result. With twelve 3U satellites, its deployer mass is **10.55 kg per customer**, versus about **6 kg** for the spring canister comparator used in the thesis. This is **1.76x** as much hardware per 3U payload and fails the preset 15% mass-parity band. The study therefore reaches a decision: the modelled orbital benefit does not make this Gen5 configuration a defensible 3U selection. The result completes the analytical comparison; it does not require a prototype to know that the modelled mass criterion failed. Physical payload loads, magnetic cleanliness, cycle life, and provider interfaces remain questions for a different evidence phase.

The later Gen6 programme retains the hosted sequential-deployment mission while reopening mechanism choice. It investigates an approximately **1-2 m/s to 1 km/s** release-speed envelope across electromagnetic, mechanical, stored-energy, gas/fluid, hybrid, BOLLEY and conventional options. The high end is a requirement to examine and may prove infeasible. The near-term goal is one named, reproducible, independently reviewable test-article configuration with a corresponding build/test release decision.

## 1. Introduction

### The deployment problem

The primary mission determines the launch vehicle's delivered orbit. Secondary CubeSats may have limited onboard propulsion, mass, power, or programmatic freedom to correct that orbit. Standard canisterised spring systems are mature and effective at clearance and separation. Their release conditions, however, are governed by spring energy, payload mass, and the device rather than a separately commanded velocity profile for every satellite.

The distinction between **release timing** and **relative-state change** is central. Waiting alone between zero-relative-impulse releases from an unchanged host does not produce persistent in-track phase spacing. Spring impulse, differential drag or a host maneuver can produce a relative state; a commanded velocity along or against orbital motion can change the satellite's orbital energy and semi-major axis. VOLLEY studies the latter capability while retaining a clear comparison with spring release and other alternatives. It does not claim that all constellation phasing requires a motor.

The host is an active participant in the mission model. After its primary mission, a suitable hosted platform would need attitude control, available resources and permission to operate. It could perform coarse orbital repositioning; VOLLEY would assign each secondary payload a local release state. No specific upper stage has been selected or approved for that role.

### Buyer and spacecraft fit

The [audited market and customer-fit report](#) checks the relevant Drive market thesis and independent business-case review against current primary sources. [Planet's FY2026 filing](#) and [Spire's 2025 filing](#) show repeatable, configurable smallsat production. [NASA's 2026 launch review](#) confirms that a primary mission can constrain the orbit available to rideshare payloads. Together, these support a *mission hypothesis*, not customer demand.

The likely party that could pay for individualized release states is a carrier, rideshare integrator or constellation operator responsible for a delivered manifest. A standardized satellite-bus maker may specify interfaces but need not buy the system. Replenishment batches and mixed-payload rideshare are candidate jobs. A one-satellite mission already in a suitable orbit is a poor fit and should favor a qualified conventional dispenser. The analyzed Gen5 system carries twelve 3U spacecraft but fails its own 3U mass screen; a PocketQube batch is a **future interface redesign**, not an accommodated or qualified Gen5 payload class. Payloads sensitive to magnetic fields

need an explicit exclusion or design remedy. No buyer, payload or provider has confirmed these requirements.

For an economic result, price and simulate the same manifest delivered with ordinary spring impulse, spring plus host manoeuvres, onboard propulsion, an orbital-transport service and VOLLEY. The published [SpaceX rideshare starting price](#) cannot stand in for an installed VOLLEY cost or savings quote. A three-to-five-year replacement cycle and a 5–50 m/s release range are research hypotheses from the Drive market report, not Gen5 operating requirements or established market demand. The Gen6 1 km/s upper objective is a different, unvalidated regime.

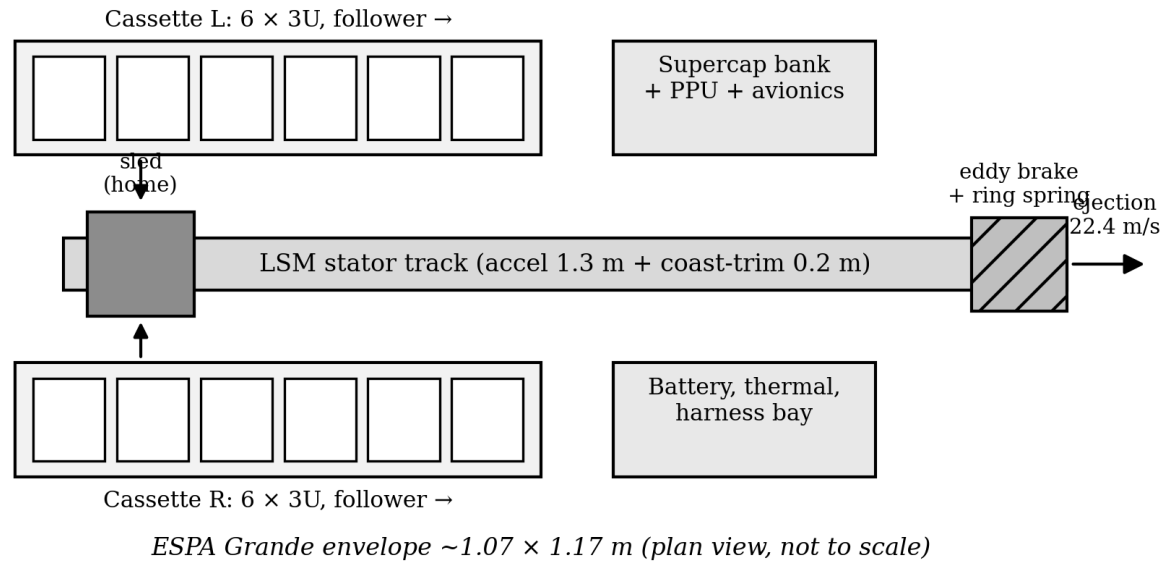


Figure 2: Gen5 layout and mission hardware elements. Diagram from the thesis analysis.

## 2. Literature review

| Approach                             | Established capability                                                        | Relevance and limit                                                                                                                                       |
|--------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spring deployers                     | Flight heritage and simple separation near the low-m/s range [1]              | Strong benchmark for reliability and installed mass; release energy is largely built into the device.                                                     |
| Spring release and differential drag | In-track spacing and gradual orbital adjustment [3]                           | A relative impulse or drag difference creates the separation; waiting alone from an unchanged co-orbital host does not.                                   |
| Orbital transfer vehicles            | Coarse orbital repositioning through a propulsive carrier                     | Valuable comparator for whole-mission delivery; greater spacecraft and integration burden must be counted.                                                |
| Electromagnetic CubeSat studies      | Magazine transport and electromagnetic launch have already been studied [4–6] | Some configurations impose conductive armatures, high acceleration or different maturity; novelty cannot be claimed for electromagnetic deployment alone. |

The most relevant high-speed electromagnetic comparison in the manuscript is Feng *et al.* [4],

which models a 321.56 m/s launch of a 20 kg CubeSat but at acceleration on the order of thousands of g and with an armature/interface issue. Zhao *et al.* [5,6] study stacked CubeSat transfer and storage, including measured transport behavior in a related mechanism. These works establish that electromagnetic delivery and magazine transport are prior art. VOLLEY's more limited question is whether a **commandable** release can be delivered to an **unmodified** satellite at a much lower acceleration while accounting for the whole installed system. No universal CubeSat quasi-static acceleration qualification is assumed; the launch provider and payload-specific limits must govern a real design [2,8].

### 3. Research gap

The analytical gap is not “no one has considered electromagnetic deployment.” It is the coupled trade between: (i) commanded release, (ii) payload-side interface and acceleration, (iii) a reusable feed/release architecture, (iv) host reaction and mission utility, and (v) installed system mass and faults. A speed-only comparison can make a concept appear attractive while omitting its sled, power bank, brake, magazine, enclosure, control, and retained mass after each shot.

The thesis addresses that gap by constructing a complete Gen5 computational configuration and testing its own claims against explicit comparisons. It also records when those tests weaken the concept. Physical viability remains a separate research gap: numerical model agreement cannot establish manufactured tolerances, actual force, payload cleanliness, cycle reliability or provider compatibility.

### 4. Objectives and outcome

| Analytical research objective     | Completed outcome                                                | Interpretation                                           |
|-----------------------------------|------------------------------------------------------------------|----------------------------------------------------------|
| Model a commandable 3U system     | Gen5 CAD, magazine, sled, motor, arrest and power architecture   | System-level design record, not manufactured hardware    |
| Quantify the release              | Rated shot, circuit and closed-loop simulations                  | Predicted performance under stated assumptions           |
| Evaluate orbital value            | Orbit model and selected independent numerical cross-checks      | Conditional energy and lifetime benefit                  |
| Test whole-system competitiveness | Mass rollup, spring/alternatives comparison and preset criteria  | Gen5 fails the 3U mass target                            |
| Challenge model credibility       | Numerical verification, defect corrections and provenance record | Scope and remaining physical uncertainties made explicit |

These objectives summarize the completed analytical research for this final-review presentation. They are a presentation synthesis, not a claim that the university approved this exact list in advance. The physical development tasks are shown later under future work.

### 5. Research methodology

The project follows an engineering evidence chain:

1. **Mission/use-case framing:** specify the payload, host responsibilities, target orbital change, safety limits and competent conventional comparators.
2. **Architecture trade:** examine force production, loading, restraint, guidance, arrest, reset and fault behavior, rather than only ideal acceleration.
3. **Geometry and installed mass:** build CAD geometry and count the sled, arrays, enclosure, feed, power, controls and other installed parts.

4. **Coupled models:** compute magnetic field and thrust, shot dynamics, power/circuit behavior, control dispersion, structural/flow responses and orbital consequences.
5. **Challenge the claims:** use alternate numerical methods and acceptance bands, preserve misses and update the baseline only with a recorded reason.
6. **Decision:** compare orbital benefit against mass, reliability, payload/host compatibility and the available alternatives.

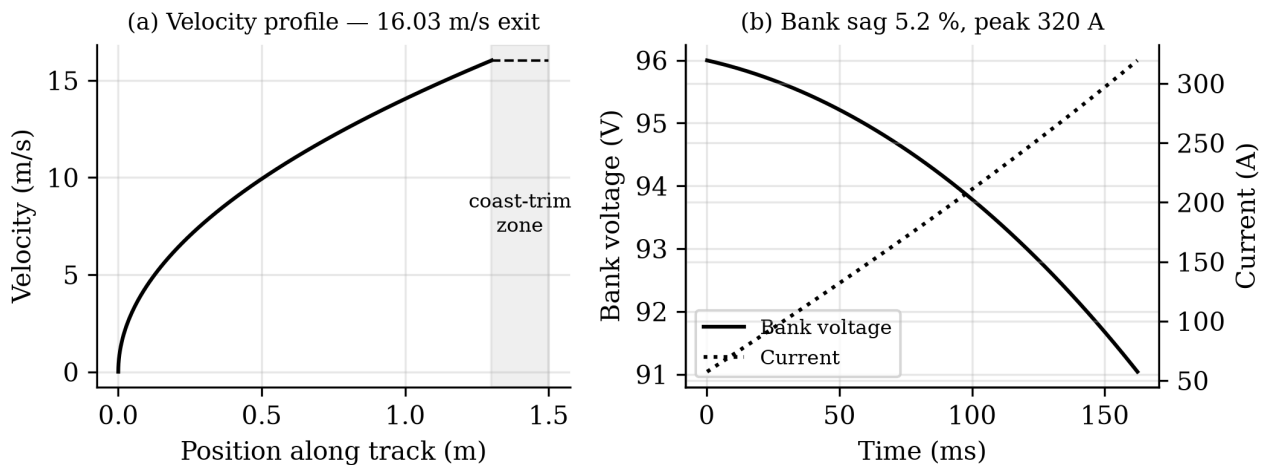


Figure 3: Modelled shot: velocity, bank voltage and current at the rated point.

## Numerical checks and their limits

The magnetic field was compared across an analytical representation, magpylib, and 2-D/3-D finite-element calculations. A 3-D midgap solve supports the field model at its tested location but does not independently measure the depth-integrated thrust constant. Shot and circuit predictions were compared with ngspice at stated assumptions. Structural calculations use CalculiX; airflow calculations use OpenFOAM. Orbital calculations include Cowell and GMAT cases. The independent orbit work rejected an earlier assertion that lifetime benefit was invariant across solar activity. Each check applies to its declared case and boundary conditions; none is a hardware test.

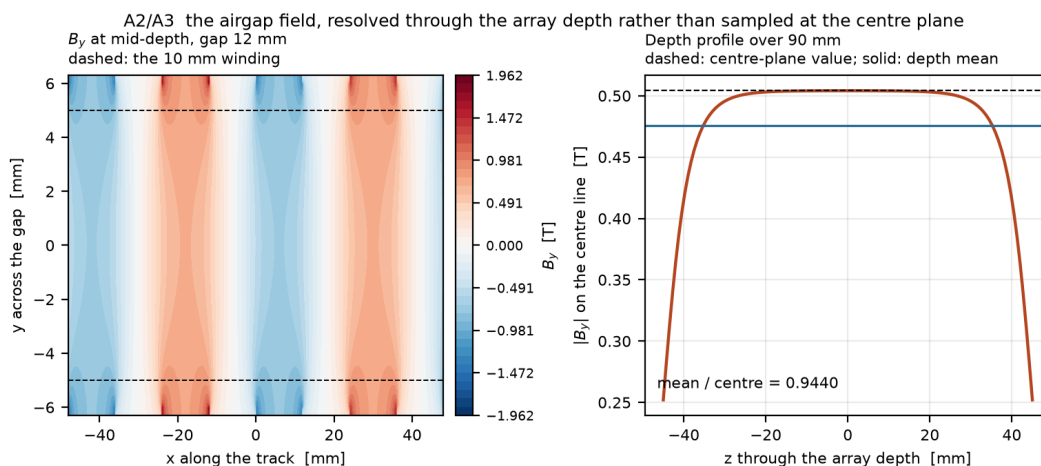


Figure 4: Representative magnetic-field model output. Source: thesis figure A02.

## 6. Timeline

### Documented evolution

| Period              | Milestone                                                    | Confidence in date                   |
|---------------------|--------------------------------------------------------------|--------------------------------------|
| 2023                | Free-flyer idea reframed around a host platform              | Documented design decision           |
| Mid-2025            | Coilgun concept shifted toward a linear synchronous motor    | Approximate historical milestone     |
| 2025-July 2026      | CAD generations and mass model developed                     | Some dates reconstructed/approximate |
| July-September 2026 | Gen5 analysis, numerical checks, defects and thesis record   | Dated repository/analysis records    |
| October 2026        | Gen5 findings and design decision presented for final review | Review package dated 10 October 2026 |

The thesis repository's reconstructed history expressly marks several early milestone dates as approximate. The analytical baseline was revised as evidence improved: a CAD-derived sled mass displaced a lighter parametric estimate; a depth-resolved thrust calculation lowered the rated speed; an enclosure buildup exposed the 3U mass penalty; and an independent orbit check rejected an overbroad solar-invariance claim. These revisions are part of the completed study, not signs that the final numbers are placeholders. Gen6 research gates are described only in §10, Future work.

## 7. Work progress

The thesis deliverables are in hand: an authored manuscript, CAD renders and model geometry, analysis scripts/results, numerical verification run sheets, figures, a baseline record, a defect register, and a provenance statement. Numerical investigations cover the motor field, shot dynamics, bank/circuit, mass, orbit propagation, separation and selected structural, airflow and reliability issues. The study closes by retaining failed criteria and superseded results rather than silently replacing them.

The deliverable here is a completed computational thesis, not a flight article. The later long gas-guide and independent spring-cell studies are historical comparators. Gen6 has a mission direction and investigation envelope but is not a result of the Gen5 thesis. Its mechanism and exact payload/provider cases remain future choices.

## 8. Results

### Gen5 configuration

The analysed system uses a double-sided Halbach-array linear synchronous motor to propel a reusable permanent-magnet sled along a 1.5 m track. Two transverse cassettes feed twelve 3U satellites. A capacitor-based bank supplies the pulse, and a brake arrests the sled after payload release. The sled mass, computed from detailed CAD solid volumes, is **9.45 kg**; this value reduced the exit speed relative to an earlier optimistic parametric estimate. The CAD render conveys the arrangement; it is not evidence that parts have been manufactured or assembled.

### Rated 3U shot

| Model output         | Rated result | Interpretation                    |
|----------------------|--------------|-----------------------------------|
| Exit velocity        | 16.029 m/s   | Modelled 3U reference shot        |
| Payload acceleration | 10.07 g      | Not a payload qualification claim |
| Pulse duration       | 162.3 ms     | Modelled motor actuation          |

| Model output                          | Rated result | Interpretation                          |
|---------------------------------------|--------------|-----------------------------------------|
| Peak current                          | 320 A        | Component implementation unverified     |
| Electrical energy drawn, gross        | 2,782 J      | Bank/circuit model                      |
| Recovered after release               | 47 J         | 39 mm available recovery zone           |
| Sled energy to brake                  | 1,162 J      | Major loss and arrest burden            |
| Electrical-to-payload efficiency, net | 18.8%        | 514 J payload kinetic energy / net draw |

The energy flow matters more than the velocity headline. The sled and its arrest hardware are the price of keeping the customer satellite unmodified. Regeneration returns only a small fraction of sled energy within the available length, so the contactless brake remains necessary. A design comparison that credits full sled recovery would misstate the system efficiency and installed burden.

## Command precision

The closed-loop Monte Carlo model predicts **0.0274 m/s ( $3\sigma$ )** exit-velocity dispersion around a **15.8 m/s** fleet setpoint. It uses assumed sensing and plant tolerances, with headroom below the open-loop ceiling. This is a simulated distribution, not a measured release repeatability.

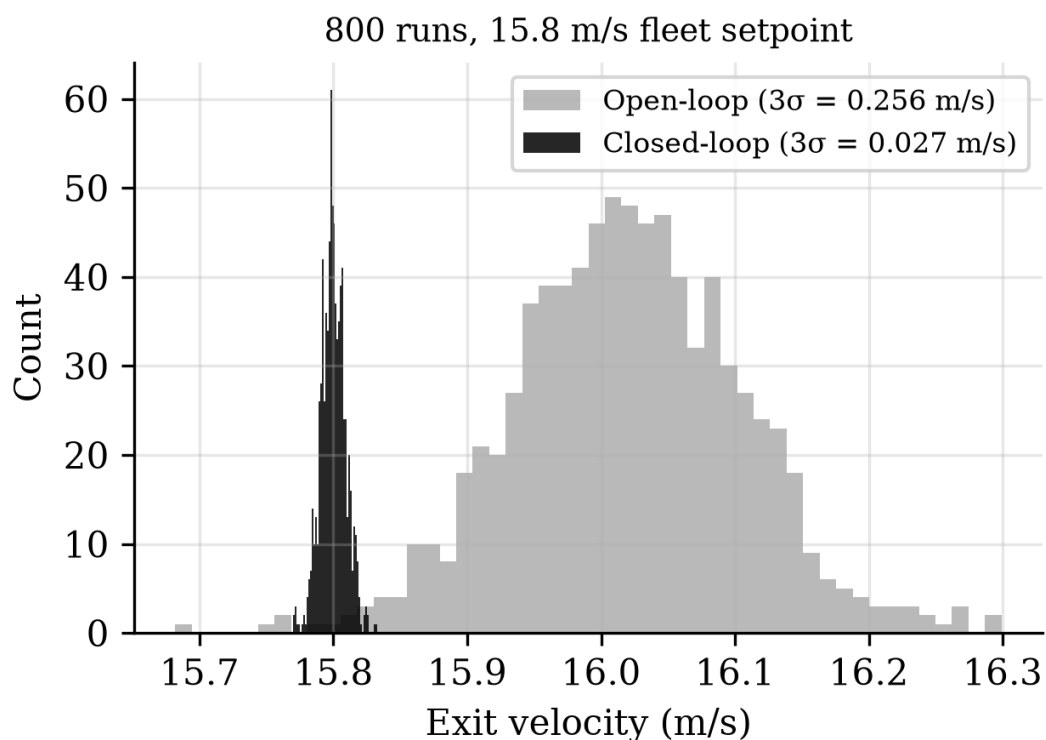


Figure 5: Closed-loop simulation distribution at the fleet setpoint.

## Orbital utility

For the stated orbit and mean-solar-activity case, one maximum-speed release predicts a **28.8 km** semi-major-axis increase and a **1.60x** lifetime multiplier for a propulsionless satellite. A separate Cartesian two-body integration of the current rated state returns **28.800775 km**, agreeing within the declared numerical band. This checks only the immediate orbital-energy result. The atmospheric lifetime multiplier has not been independently reproduced at the exact current point. Timing alone, with zero relative impulse and no host or drag difference, produces neither persistent phasing nor this semi-major-axis change.

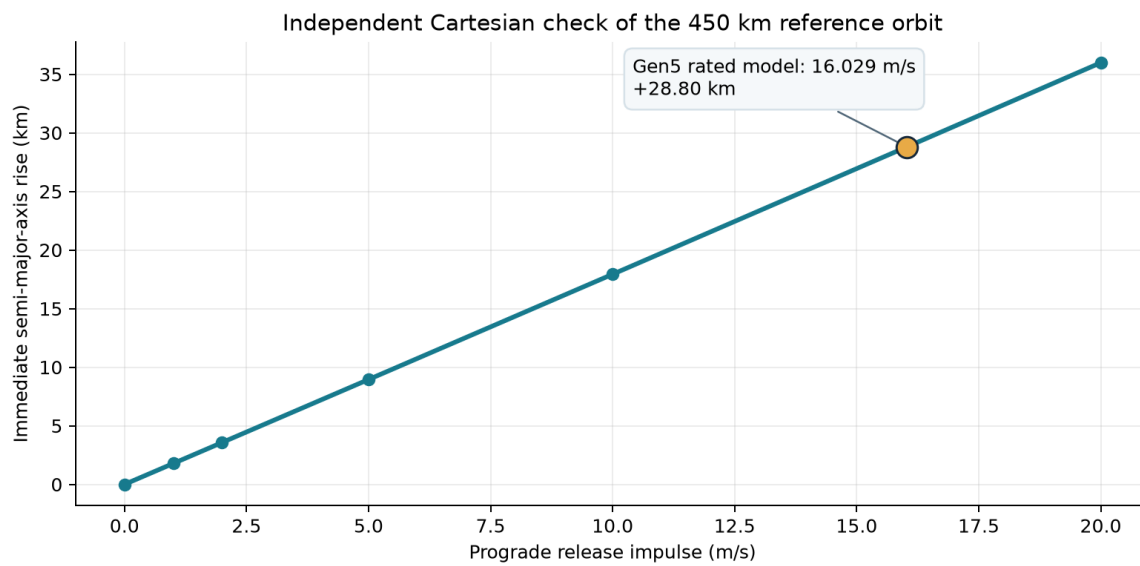


Figure 6: Independent rated-state two-body orbit check. The lifetime claim is outside its scope.

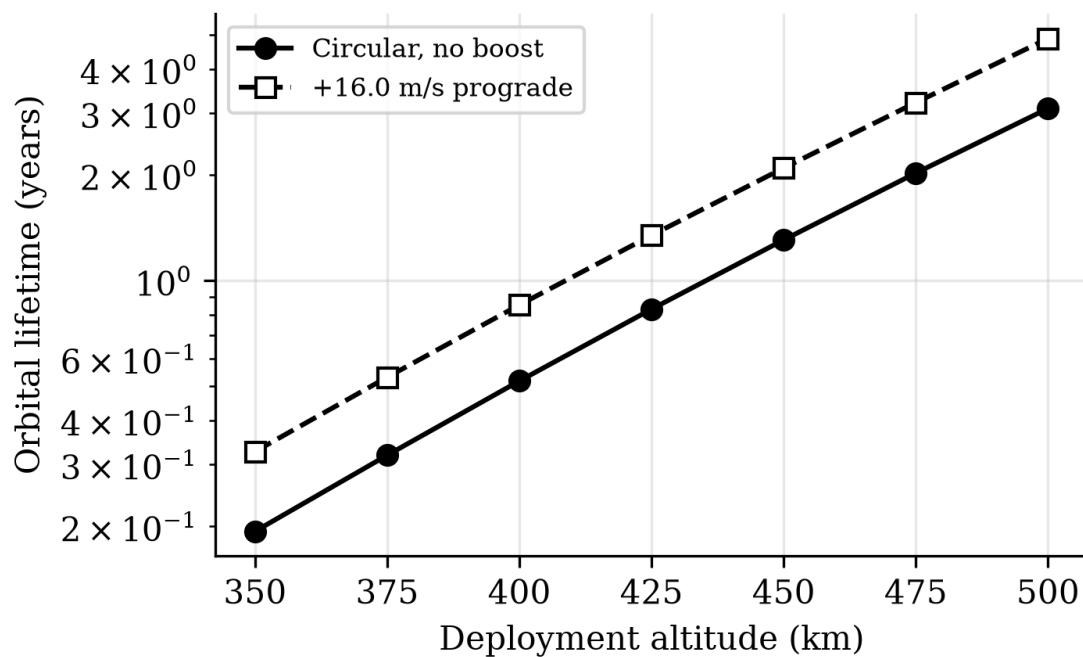


Figure 7: Modelled orbital-lifetime response. Absolute values depend on atmosphere.

## Installed mass changes the decision

The modeled Gen5 dry rollup is **126.6 kg** and loaded mass **174.6 kg**; these include assumed hardware and historical Gen3 sled volume, not a mass extraction from a complete installed Gen5 assembly. At twelve 3U customers, the deployer weighs **10.55 kg per customer**, compared with approximately **6 kg** for the spring canister comparator used in the analysis. The ratio is **1.758**; the preset acceptance band required parity within 15%. That band fails by a large margin. Replacing an **8 kg enclosure placeholder** with a **50.04 kg** geometry-derived buildup exposed much of the penalty. Even changing structure alone does not recover parity in the studied options. The result is a reason to reopen the architecture, not to conceal the comparison.

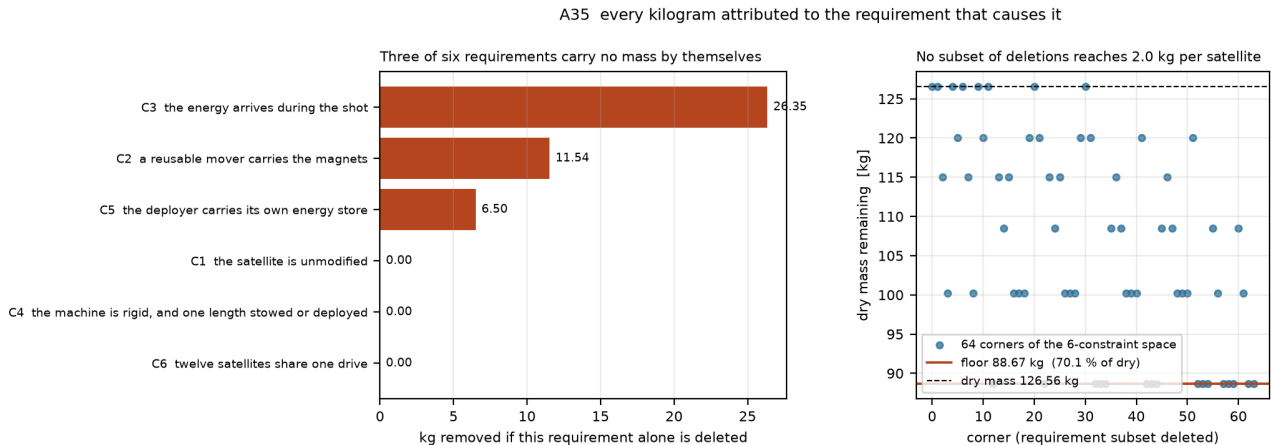


Figure 8: Constraint and mass-attribution analysis; mass is a model rollup.

## Native CAD and packaging review

The geometry package now includes a FreeCAD 1.0 native document, eight re-exported STEP parts, and a 20-instance STEP assembly. Each imported solid passed FreeCAD validity and nonzero-volume checks. The source solids were produced in CadQuery; importing them into FreeCAD does not create parametric feature history or prove that the assembly can be manufactured. The [CAD review report](#) identifies source revisions and checks, while [P116](#) records the numerical intersection result.

The side-fed reference assembly has **526 mm** clear enclosure width. Its **205 mm** track plus two **166 mm** cassettes require **537 mm**, an **11 mm** shortfall even before running clearance. The exact solids overlap by **32,915 mm<sup>3</sup> per side** in this placement. This fails that packaging arrangement. A different feed architecture, enclosure or dimension set requires a new configuration and reassessment; the current render must not be presented as an assembled flight article.

## Decision and evidence boundary

The analytical result has four parts. The model supports a commandable release and conditional orbital-energy benefit; the full-system 3U mass criterion fails; the stated side-fed layout fails its CAD width check; and physical behavior remains unverified. The latter includes release-cycle reliability, shock and arrest loads transmitted to stowed satellites, field exposure near a customer payload, host attitude/control authority and provider approval. The known-problems register ranks several as potentially design-fatal. These findings limit product claims and define the research conclusion.

## 9. Conclusion

The Gen5 computational thesis reaches a research conclusion within its stated scope. Its model predicts a commandable moderate-acceleration release and a conditional orbital-energy benefit, with a separate check of the immediate two-body orbital change. Installed-mass accounting misses its preset 3U benchmark, and the side-fed reference CAD assembly fails its width check.

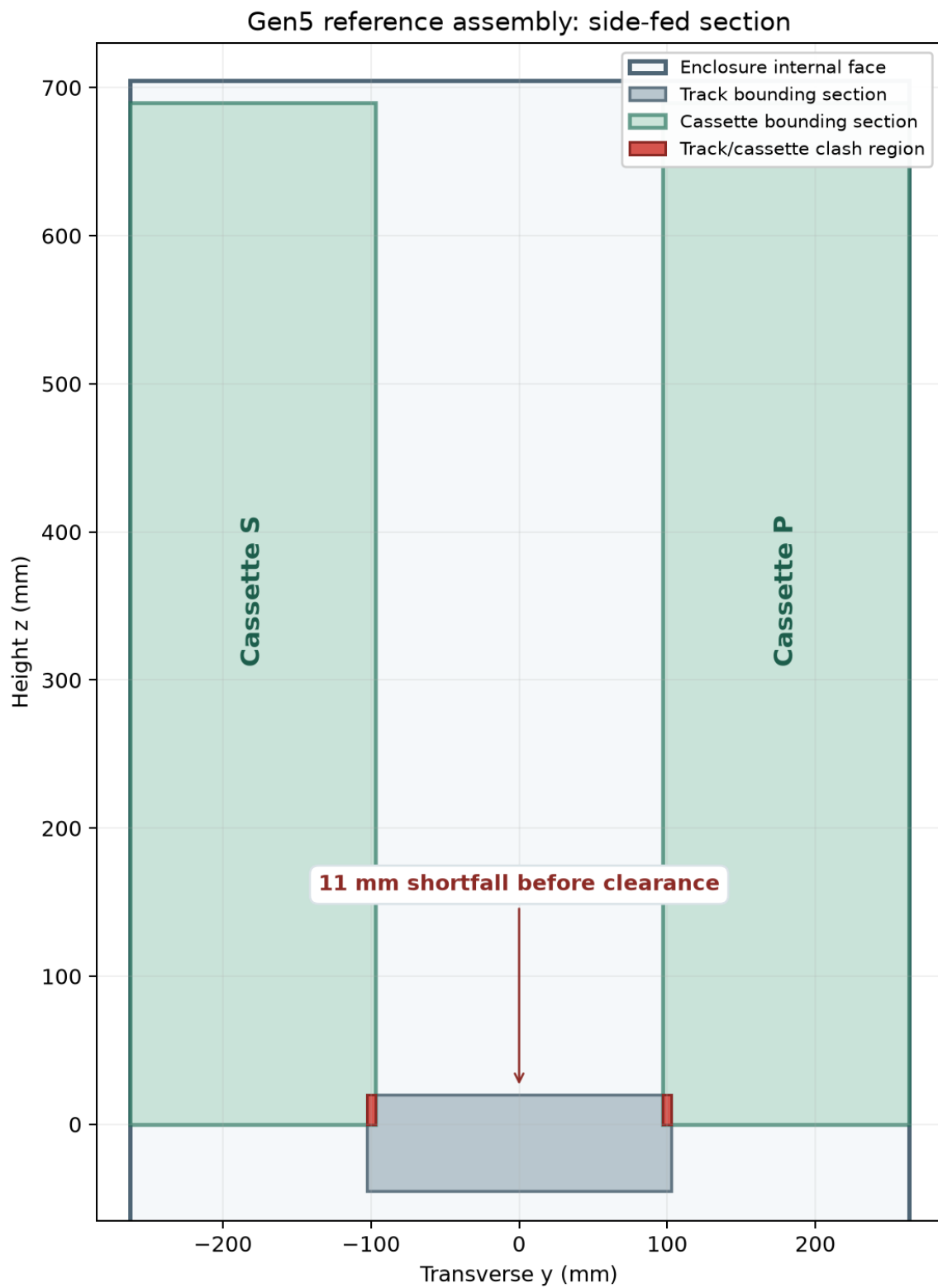


Figure 9: Dimensioned side-fed reference cross-section; this configuration fails the width check.

The current configuration is therefore not a selected flight baseline. Hardware behavior and interfaces remain outside the tested evidence.

The thesis’s strongest engineering contribution is the transparent decision record: the full machine was modelled, independent numerical checks corrected several claims, and an unfavourable mass result was retained. That closes the Gen5 study and provides a defensible input to a separate future architecture trade.

## 10. Future work

Gen6’s controlling objective is a reusable launch path that sequentially accepts different supported CubeSat sizes and commands each departure condition. The approximately 1–2 m/s to 1 km/s span is an investigation envelope; the upper end may force acceleration, energy, travel, payload or provider constraints that reject candidates. Electromagnetic systems are preferred for comparison but are not preselected. Competent springs, electromechanical/stored-energy, gas/fluid, hybrid, BOLLEY and conventional mission substitutes belong in the same comparison boundary.

The near-term milestone is one named test-article configuration that can be independently reviewed. Its package needs controlled geometry and drawings, a bill of materials, host and payload interfaces, full installed mass/energy/thermal accounting, assembly and inspection instructions, instrumented procedures, predetermined pass/fail and stop criteria, uncertainty and model-credibility evidence, retained failures, and an explicit build/test release decision. A rejected candidate is a valid research outcome; it must not be presented as a build-ready survivor.

## 11. References and project records

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10. A. Mishra, *VOLLEY-thesis*: authored manuscript, baseline, provenance, figures and validation records, repository head b30ffdc, 28 September 2026. Project record, not independent experimental evidence.
11. VOLLEY/BOLLEY, *Current Plan and Status — PLAN-R2*, controlled working baseline, 29 September 2026. Project plan, not achieved performance.

## Appendix A. Numerical and CAD audit trail

| Result                         | Reproducible record                                                                              | What remains outside the check                   |
|--------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Rated shot and energy          | <a href="#">analysis/results/motor_results.json</a> ,<br><a href="#">analysis/motor_model.py</a> | Unmeasured force and actual circuit parasitics   |
| Rated immediate orbital change | P115,<br><a href="#">rated_orbit_independent.py</a>                                              | Atmospheric lifetime and controlled host mission |

| Result                        | Reproducible record                                                                               | What remains outside the check                                 |
|-------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 3U mass decision              | <a href="#">mass_properties.json</a> , <a href="#">BASELINE</a>                                   | Hardware weighing, complete installed host burden              |
| Native geometry and collision | <a href="#">Gen5_Review.FCStd</a> ,<br><a href="#">FREECAD_EXPORT.json</a> , <a href="#">P116</a> | Tolerance, feed motion, payload accommodation and provider fit |
| Wider model and open gates    | <a href="#">computational review</a> , <a href="#">CAD review</a> , <a href="#">open problems</a> | Final verification closure and physical development            |

The run sheets report software, inputs, declared comparison bands and files. Figures are generated model or CAD visualizations. A screenshot of a solver interface would not by itself establish a correct result; the editable model, data and case boundary are the audit trail.

## Appendix B. Submission and freeze status

The project is ready to present as a computational study with recorded adverse findings. It is **not ready** to be described as a fully validated design or released product. Before an academic freeze, close or explicitly disposition the decisive electromagnetic, feed/release, brake, host, structural and installed-system questions in the local evidence register; choose the exact college report template; record the final authorship, guide approval and submitted PDF checksum. Physical bench and qualification tests remain future work and must stay marked unrun.